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Artificial Intelligence Based Optimization of Network Traffic at Telecommunication Companies

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***Abstract*—With the advent of smartphones, cloud computing, and the use of many bandwidth-consuming applications, the demand for traffic in telecommunication systems has increased manifold. Therefore, the effective management of the traffic in telecommunication systems is of prime importance. Artificial Intelligence and Machine Learning techniques provide effective solutions for the modeling and prediction of complex traffic in telecommunication systems. This paper presents an evaluation of Linear Regression Model, Random Forest Model, Support Vector Machine Model, Long Short-Term Memory(LSTM) Model, and the Transformer model for the prediction of traffic in telecommunication systems. The performance of the mentioned models is evaluated using the RMSE, MAE, and R^2 metrics. The evaluation demonstrates the effectiveness of ensemble and deep learning(DL) techniques in improving the accuracy of the prediction of the traffic in telecommunication systems.**

***Index Terms*—Artificial Intelligence(AI), Machine Learning techniques, Network Traffic Prediction, Telecommunication Systems, Traffic Optimization**

I. Introduction

The telecom industry has witnessed tremendous change in the last few years due to the proliferation of mobile networks, cloud computing, and digital technologies [1].As the user base grows in the telecom industry, managing the massive amount of daily data transfer also becomes a challenge. For this purpose, artificial intelligence technologies have been used to improve the overall efficiency in the telecom industry. AI-based solutions have helped telecom providers to monitor the network traffic and predict future demand [2].By using big data analytics along with machine learning algorithms, telecom companies have improved their operational efficiency and reduced infrastructure costs [3]. Predictive models have also been used to improve the overall efficiency in the telecom industry [2].

Network traffic usually varies over time, location, and user behavior. Machine learning algorithms have been used to predict the variation in network traffic [4].These models have helped telecom providers to improve their overall efficiency by using network traffic data to identify patterns that were previously unknown to the network provider [3]. From a mathematical perspective, network traffic prediction can be represented as

$$T_{t+1} = f(T_t, T_{t-1}, \ldots, T_{t-n}) \tag{1}$$

where Tt+1 = predicted traffic value Tt = current traffic value n = number of historical observations used for prediction. This formulation allows machine learning algorithms to learn the relationship between past traffic values and future network demand.

II. Related Work

Significant progress has been achieved in the application of the ML techniques for the telecommunication networks, as discussed in the literature [2]. The DL techniques, namely Long Short-Term Memory (LSTM), are best suited for the time series prediction, as discussed in the literature [6]. Another model is the transformer, which also uses the self-attention mechanism to handle the temporal dependencies. Thus, the application of the artificial intelligence techniques is vital for the 5G and future networks, as discussed in the literature [8], [9]. But the prediction of the traffic is challenging due to the varying nature of the users and the infrastructure [10].

Researchers have also applied anomaly detection algorithms for the identification of unusual network behavior and possible cyber attacks in telecommunication networks [4]. Researchers have also proposed a variety of traffic classification algorithms using Support Vector Machines and K-Nearest Neighbor algorithms for the improvement of network

performance and effective management of network congestion. Artificial Neural Networks have also shown significant potential for the effective learning of complex patterns present in telecommunication network traffic data [4]. Similarly, (LSTM) networks have been widely applied for the effective management of network traffic using the predictive approach. The ability of the LSTM network to capture complex patterns present in network traffic data is significant.

Contemporary research on the application of artificial intelligence algorithms has also been conducted on the application of AI algorithms for the management of next-generation communication systems such as 5G networks. This indicates the importance of the application of intelligent systems for the optimization of network performance with the help of intelligent systems capable of automatically optimizing network performance [6]. Despite the effective application of AI algorithms for the management of network traffic, the effective prediction of network traffic is still a complex task due to the fluctuating behavior of network usage.

III. Proposed Methodology

A. Data Preprocessing

Data cleaning, feature extraction, encoding, and normalization are essential preprocessing steps [7] for improving model generalization.

B. Min-Max Normalization

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{2}$$

IV. Machine Learning Models

A. Linear Regression

Linear Regression models linear relationships between predictors and outputs [8].

B. Random Forest

Random Forest enhances prediction stability using ensemble learning and variance reduction [9].

C. Support Vector Machine

$$f(x) = w^T x + b \tag{3}$$

Support Vector Machines maximize margin separation in high-dimensional spaces [10].

D. LSTM Network

LSTM networks capture long-term sequential dependencies through memory gating mechanisms [6].

E. LSTM Gates

Forget Gate

The forget gate decides which information needs to be removed from the previous state of the cell [6]. In time series prediction, not all information is equally important in determining the future forecast. The forget gate removes any irrelevant information using a sigmoid activation function. The output of this function varies between 0 and 1. When the output is close to 0, it means the information should be removed, while when the output is close to 1, it means the information should be kept selectively.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{4}$$

Input Gate

The input gate controls the information being stored in the memory cell [6]. It checks the level of importance of the current information being entered into the memory cell, comparing it with the previously stored hidden information. This enables the model to be sensitive to the dynamic patterns of the telecom data, such as the sudden spikes during peak usage times or the seasonal changes in usage patterns. This gate regulates the amount of new information being entered into the memory cell.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{5}$$

Output Gate

The output gate is what passes on the last word of the LSTM cell at a given time [6]. This gate determines what parts of the updated cell state will reach the next layer. This gate is what drives the prediction, all while preserving the critical long-term patterns. In network traffic forecasting, the output gate helps the model narrow in on actual traffic trends and reduces the influence of noise.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{6}$$

These gates collectively enable the LSTM network to retain significant traffic patterns over time while discarding irrelevant information.

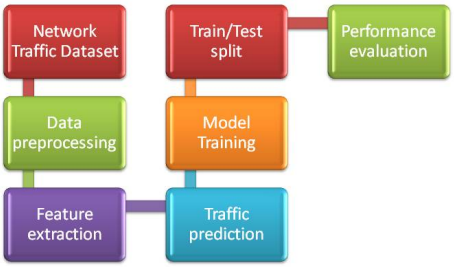

Fig. 1. Diagrams of how machine learning models are processed.

F. Transformer Model

Transformer architecture uses multi-head self-attention to model extended contextual relationships [11].

V. System Workflow

Figure 1 presents the structure of the proposed Cell-Aware Temporal Transformer (CATT). The input layer receives the temporal features including hour, day, and month, and the cell tower ID. This information is processed in a cell embedding layer to identify unique patterns related to cell towers. The transformer encoder processes the information to identify the pattern of changing traffic over time [12]. This is achieved through the application of attention to identify long-range relationships. A final fully connected layer is used to generate the predicted traffic value.

Figure 2 presents the workflow of the proposed network traffic prediction system. The process begins with dataset collection from telecom network logs. The data is then preprocessed to remove inconsistencies and extract relevant features such as day, month, and cell identifiers. Once the preprocessing is done,the dataset is split into a training set and a testing set.The machine learning and deep learning models aretrained on the training set, and predictions are made on the testing set. At last, the model's performance is checked using standard error metrics like MAE method, RMSE method, and R^2 method.

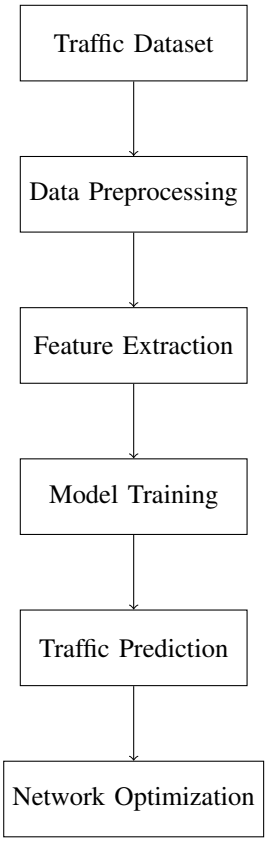

Fig. 2. Workflow of AI-based network traffic prediction system

A. Dataset Description

The research uses a data set based on historical network traffic collected through telecom towers. Each entry has a list of attributes that describe the data.

- Date
- Hour
- Cell Identifier
- Traffic Value

This data set tracks the patterns of traffic movement over various periods of time and space. By structuring the data in a particular format, it assists in the identification of patterns in network demand, which in turn helps in building predictive models to forecast the patterns of traffic movement in the future.

VI. Experimental Setup

The experiment is implemented using Python and the following frameworks: Scikit-Learn and TensorFlow [15]. The Transformer models are effective in learning long-range dependencies in the traffic

data [7]. Spatial-temporal feature integration helps improve the robustness of the model [4].

A. Performance Metrics

It is also evident from Figure 2 that the model with the lowest value of RMSE is the Random Forest model, and this is a good measure of the performance of ensemble models [13]. Another model, called the Transformer, is also performing well, and this is due to the attention mechanism of this model [7]"Mean Absolute Error (MAE) calculates, on an average basis, how far the model's predictions are from actual values" [8]. If the value of MAE is low, it implies that the predicted values are closer to the actual observations, and hence the performance of the model is good [4].

$$MAE = \frac{1}{n}\sum |y_i - \hat{y}_i| \tag{7}$$

[26] Where , yi = actual value , yi = predicted value ,n = number of observations.

In the prediction of telecom traffic, MAE indicates how close the predicted traffic values are to the real traffic measurements of the network.

The Root Mean Square Error is defined as "the square root of the mean of the squared differences between the predicted values and the actual outcomes" [8]. The larger the prediction error, the more it affects the RMSE because of squaringref14, making it more appropriate to use RMSE if larger errors should be penalized more.

$$RMSE = \sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2} \tag{8}$$

The R-squared value indicates the goodness of fit of the actual values with the predicted values [27]. It also illustrates the proportion of the variation in the dependent variable that is explained by the independent variables. [14].

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \tag{9}$$

Figure 1 presents a comparative analysis of the RMSE scores of the evaluated machine learning models. It is true that the Random Forest model has the lowest prediction error among the other models [9]. However, the Transformer model also demonstrates its competitiveness with the other

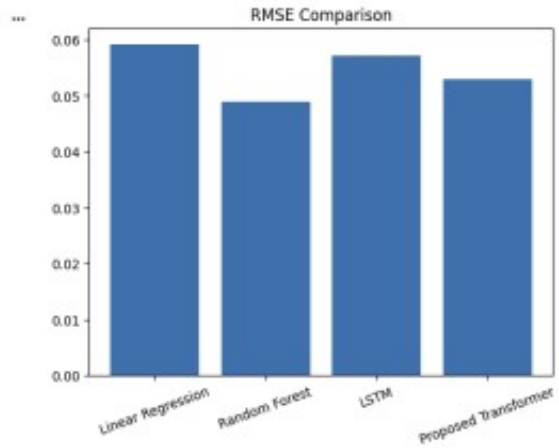

Fig. 3. Comparing RMSE for different machine learning models.

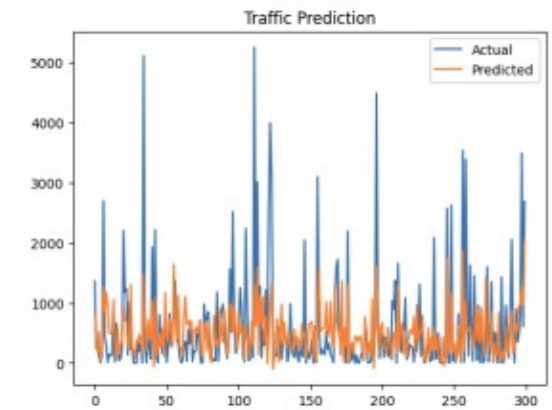

Fig. 4. Comparison between actual and predicted network traffic values.

models [4].On the other hand, the Linear Regression model demonstrates a higher error level because it is not capable of handling the complex nonlinear patternsthat are present in the telecom network data. Thus, the analysis indicates the superiority of using advanced machine learning models for network traffic predicting accuracy [4].

Figure 4 presents a visual of the arrangements between the model's predictions and the actual values of the traffic. This aligns with the overall pattern of the actual traffic. This indicates that the model can learn the temporal patterns in the provided telecom data [3]. There are small deviations during the abrupt changes in the traffic levels; however, the model has been able to learn and get the general pattern of the network traffic [4].

TABLE I
RESULTS OF MACHINE LEARNING MODEL PERFORMANCE

Model	RMSE	MAE	R^2	Accuracy (%)
Linear Regression	0.059120	0.039510	0.172670	94.09
Random Forest	0.048860	0.031324	0.434916	95.11
LSTM	0.057083	0.039038	0.228715	94.29
Transformer	0.052888	0.035051	0.337919	94.71

VII. RESULTS

The table compares the performance of the baseline models with the proposed Cell-Aware Temporal Transformer (CATT) model. The table indicates that, in general, traditional methods such as Linear Regression have higher prediction error rates because they cannot capture non-linear patterns in the data. The Random Forest method performs a little better because it averages the results of many decision trees, allowing it to learn more complex relationships in the data [10]. However, the proposed CATT model performs better in terms of accuracy using the attention mechanism of the transformer model [4]. This shows that deep learning models are better suited to learn the temporal and spatial patterns of telecom network traffic.

VIII. CONCLUSION

This research focused on the application of artificial intelligence methods for the optimization of telecom networks' traffic. In this research, the author has used different machine learning models for the prediction of telecom traffic data and has determined the potential of these models for the prediction of future patterns. From the research, it has been concluded that the application of artificial intelligence methods has the potential for the optimization of telecom networks, resulting in the better use of resources and better quality of service.

IX. FUTURE DIRECTIONS

Future research may explore AI integration with 5G and beyond-5G systems, including network slicing and intelligent orchestration [8], [9].

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